

Homework for TDS: Problem 5

Problem Description

We consider a point particle with electric charge e and mass m confined to a one-dimensional potential well of width a , bounded by walls at $x = 0$ and $x = a$. The particle moves freely between the walls and undergoes elastic reflections upon colliding with them.

The system is driven by a spatially homogeneous, time-periodic electric field $E(t)$ with period τ . The field is piecewise constant and alternates sign every half-period,

$$E(t) = \begin{cases} +E_0, & 0 < t < \frac{\tau}{2}, \\ -E_0, & \frac{\tau}{2} < t < \tau \end{cases} \quad (1)$$

Thus, the particle is alternately accelerated to the right and to the left.

The goal is to derive a stroboscopic map describing the motion and to investigate the dynamics by constructing phase-space portraits for different values of the dimensionless parameters.

Hamiltonian Formulation

The Hamiltonian of the system is

$$H(x, p, t) = \frac{p^2}{2m} + exE(t), \quad (2)$$

where x is the position and p the canonical momentum.

Hamilton's equations yield

$$\dot{x} = \frac{p}{m}, \quad \dot{p} = -eE(t). \quad (3)$$

Since $E(t)$ is piecewise constant, the particle undergoes motion with constant acceleration during each half-period.

Dimensionless Variables

To reduce the number of parameters, we introduce dimensionless variables

$$\hat{x} = \frac{x}{a}, \quad \hat{p} = \frac{\tau p}{am}, \quad s = \frac{t}{\tau}. \quad (4)$$

In these units, one full period corresponds to $s = 1$, and the well is given by $\hat{x} \in [0, 1]$.

We also introduce a dimensionless field strength parameter

$$k = \frac{eE_0\tau^2}{am}. \quad (5)$$

The equations of motion become

$$\frac{d\hat{x}}{ds} = \hat{p}, \quad \frac{d\hat{p}}{ds} = \begin{cases} -k, & 0 < s < \frac{1}{2}, \\ +k, & \frac{1}{2} < s < 1. \end{cases} \quad (6)$$

1

Elastic Reflections at the Walls

When the particle reaches a wall, it undergoes an elastic collision. In dimensionless variables this corresponds to

$$\hat{x} = 0 \text{ or } 1 \Rightarrow \hat{p} \rightarrow -\hat{p}. \quad (7)$$

Thus, the motion consists of free acceleration segments separated by instantaneous momentum reversals.

Numerical Method

The dynamics were simulated using an exact time-integration scheme:

1. Start from $(\hat{x}, \hat{p})$ at time s ,
2. Determine the time to the next event:
 - collision with a wall,
 - field reversal,
 - end of the current period.
3. Advance the trajectory analytically under constant acceleration to that event.
4. Update the state:
 - reverse momentum at a wall,
 - change the sign of acceleration at a field switch.
5. Record the state at $s = n$.

Multiple trajectories with different initial momenta $\hat{p}_0$ were simulated for the same initial position $\hat{x}_0$. The phase-space portraits were obtained by plotting all sampled points $(\hat{x}_n, \hat{p}_n)$.

Results

The resulting phase diagrams show that the motion of the particle strongly depends on the value of the parameter k , which represents the strength of the electric field.

- Small values of k : the motion appears quite regular. The particle follows smooth and predictable trajectories, indicating that the external field only weakly perturbs the motion inside the well.
- Intermediate values of k : the behavior becomes more complicated. Some trajectories remain regular, while others start to spread out in phase space.
- Large values of k : the motion appears irregular and disordered. The particle explores a wide range of positions and momenta, and the trajectories no longer follow simple patterns.

Numerical simulations reveal a transition from regular to chaotic dynamics as the field strength parameter k increases. The phase-space portraits show stable structures and chaotic regions, demonstrating the complex behavior induced by time-periodic driving in a bounded Hamiltonian system.





